

How the Decision Layer Actually Makes Decisions

SmartLoad does not select a single algorithm for routing. Instead, the Decision Layer acts as an **controller** that merges the outputs of several sub-engines: classical load balancing, reinforcement learning, anomaly detection, and forecasting. Each sub-engine contributes a different signal, and the final routing or scaling action is computed from their combined output through a unified control bus.

1. Why Multiple Sub-Engines Exist

Each engine is responsible for a specific type of intelligence:

- **Classical load balancing** provides deterministic, safe, and low-latency rules.
- **RL Engine** learns adaptive routing policies based on telemetry.
- **Anomaly Detector** identifies nodes that are slow, unstable, or unhealthy.
- **The forecaster** predicts future load and informs proactive scaling.

These engines run in parallel, consuming the same metrics but producing different decision signals. The system does not “switch” between them; instead, it integrates their insights to produce a more robust, adaptive outcome.

2. Control-Bus Based Merging

Outputs from all decision modules are published to a shared internal channel (*control bus*). Examples include:

- RL routing scores
- classical algorithm baselines
- anomaly flags (healthy / degraded / quarantined)
- forecasted load signals for scaling
- policy constraints (from the Policy Manager)

The load balancer and autoscaler subscribe to this bus. This design allows each module to evolve independently while ensuring the Decision Layer always has a complete, consistent picture of system health.

3. Final Routing Decision Logic

The load balancer computes the final server selection using the following hierarchy:

1. Apply **anomaly detector constraints**: remove or down-weight unhealthy nodes.
2. Use **RL routing scores** for optimal adaptive placement.
3. Fall back to **classical algorithms** if:
 - RL model is still warming up,
 - telemetry is insufficient,
 - the Policy Manager enforces safe mode.

Thus, the classical engine is always available as a deterministic safety net, while the RL engine drives optimization whenever conditions allow.

5. Interaction with Forecasting and Autoscaling

Forecasting is not used for routing, but it is a crucial part of decision intelligence. It provides time-series predictions such as:

- upcoming traffic surges,
- expected saturation windows,
- low-load intervals for scaling down.

The autoscaler consumes these signals and initiates proactive scaling actions. Newly provisioned instances register with the load balancer, completing the feedback loop.

6. Summary of the Decision Flow

1. Telemetry produces real-time metrics.
2. Each sub-engine (RL, classical LB, anomaly, forecasting) computes its signals.
3. All outputs are published to the control bus.
4. The load balancer merges routing scores and applies anomaly constraints.
5. The autoscaler uses forecasted load and policy rules to scale resources.
6. New telemetry reflects the system state after the decision.

Rather than a single “algorithm selector,” the Decision Layer functions as a coordinated, multi-intelligence controller. This ensemble approach ensures stability, adaptiveness, and robustness under fluctuating workloads.